

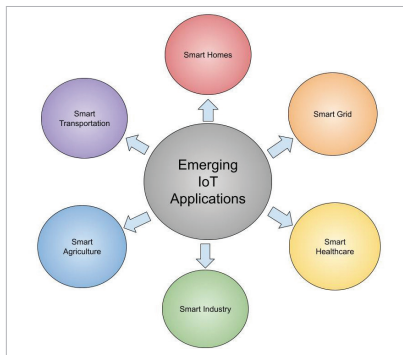


Figure 1: Emerging IoT applications

it. In many IoT applications, where reliable and real-time data delivery is required, cloud computing based services are not suitable to meet their demands. Most IoT applications face the following challenges:

- Scalability
- Reliability
- Transmission delay
- Interoperability
- Availability
- Security and privacy

The rapid growth in IoT devices has opened up several applications in different domains, and especially in the field of communications. Many of these emerging applications are open to different sets of goals and requirements, but play a vital role in making our everyday activities easy, smart, and safe. Figure 1 shows the various emerging IoT applications.

Role of edge computing in IoT

Edge computing processes and stores the data received at the edge of the network, which is near to the end users. The processing of data closer to the end devices decreases the transmission time, lowers bandwidth needs and increases the battery life of IoT devices. Edge computing offers benefits for many IoT applications, including reduced response time and improved communications efficiency, as compared to cloud computing.

IoT applications can be classified with respect to their delay sensitivity — a particular application may

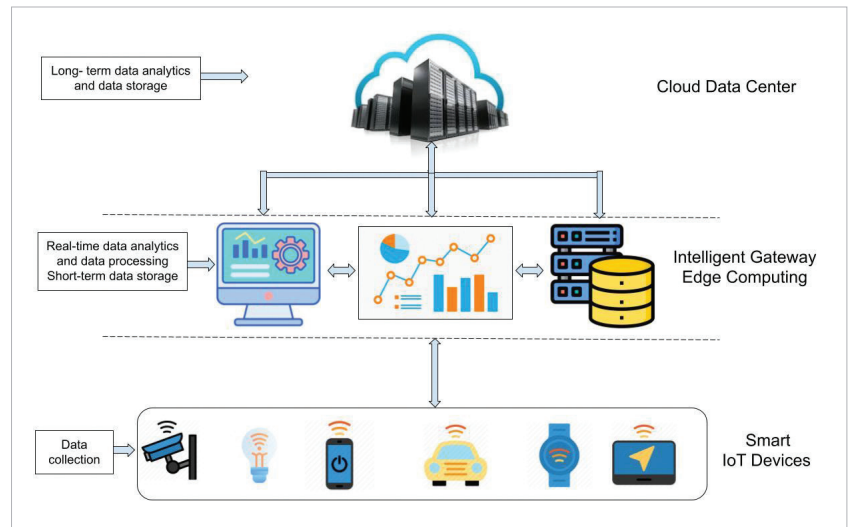


Figure 2: Edge computing-IoT layered architecture

or may not be sensitive to delay, depending on the type of parameters it manages. We can choose between cloud computing and edge computing based on this delay sensitivity. Edge computing performs well in comparison with cloud computing for IoT, though it has limited computational power and storage. Most IoT applications require real-time services rather than greater computational power and storage.

Edge computing with IoT is extensively used in many industrial applications; it enables numerous new services for enterprises and users. Many emerging IoT applications, such as smart vehicles, smart grids, depend on edge computing. Advantages of edge computing in IoT include:

- Low latency
- Longer battery life for IoT devices
- Low bandwidth requirement
- Resilience in systems
- Scalability

Fusion of edge computing with IoT

Edge computing is a cost-effective framework for extending the functionalities of cloud computing. In real-time IoT applications, it reduces the latency and also the response time.

The features of edge computing such as low latency, mobility provision, location awareness, and dense distribution can solve the challenges the IoT industry is facing.

Figure 2 shows the three-layer architecture of edge computing based IoT. The bottom layer contains various smart IoT devices, which generate the data. Data gathered from this layer is sent to the edge layer, where data processing and data analytics tasks are performed at the edge nodes to make real-time decisions. Based on the decisions taken at the edge layer, data is sent to cloud servers for exhaustive analysis and stored there.

Emulators and simulators for edge IoT computing

The excellence of any IoT system is usually evaluated by the degree to which it fulfils the requirements of its users. The performance of an IoT application can be measured with various quality metrics like latency, energy efficiency, and scalability. These measurement metrics vary from application to application, because every application has different requirements. The simulation of real-world IoT infrastructures has gained substantial interest in the research